IEOR 221

Homework 1

Module 2 (September 8, 2025)

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1) What is the monthly payment for a 5 year car loan of $20,000 with 10% fixed interest rate (monthly compounding)? Assume 5 x 12 = 60 equal monthly payments.

2) Suppose that the 6-month, 12-month, 18-month, and 24-month zero rates are 5%, 6%, 6.5%, and 7%, respectively, with semiannual compounding. What is the 2-year par yield? (The par yield for a certain bond maturity is the coupon rate that causes the bond price to equal its par value (= the notional value.). Assume the bond pays coupon semi-annually. (Hull 4.22)

3) Suppose that risk-free zero interest rates with continuous compounding are as follows:

Maturity (years)	Rate (% per annum)
1	2
2	3
3	3.7
4	4.2
5	4.5

Calculate forward interest rates for the second, third, fourth, and fifth years. (Hull 4.23)

4) The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of $4 every 6 months currently sells for $94.84. A 2-year Treasury bond that will pay coupons of $5 every 6 months currently sells for $97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year Treasury zero rates. (Hull 4.29)

5) An interest rate is quoted as 5% per annum with semiannual compounding. What is the equivalent rate with (a) annual compounding, (b) monthly compounding, and (c) continuous compounding. (Hull 4.34)